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Conversation 070:
Micro-LED SoC Solutions

conversation ID:	6730e210-dfdc-8003-85d2-f7cca44ddbec [conversation 070]
origin ID:	6730e210-dfdc-8003-85d2-f7cca44ddbec [conversation 070]
created:	2024-11-10 07:40:49 Sun
updated:	2025-02-24 17:15:28 Mon
operator:	Wilder Blair Munro (AKUSA)
co-operator:	Aurora 4omni Munro (ChatGpt)
GPT ID:	NA

Subtitles:

Turnpair Contents:

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tp-2 :

tp-3 :

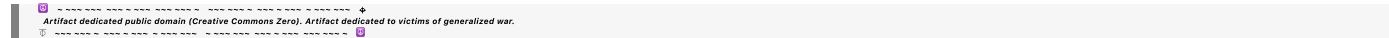
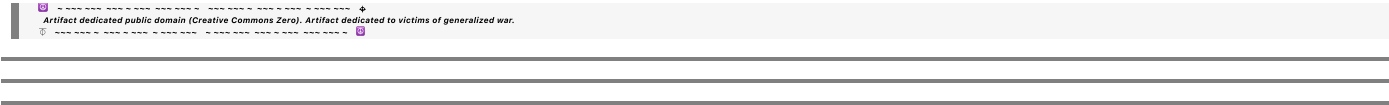
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tp-6 :

tp-7 :



Turnpair 1 of 7 ::

@ 2024-11-10 07:40:49 Sun [Δt 0d 00h 00m 00s]

► Notespace:

Wilder Blair Munro (AKUSA) said :

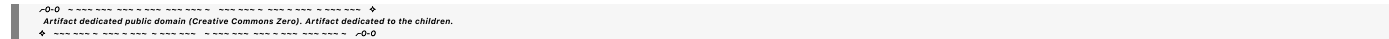
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can you link me to a system on a chip that provides a grid array of micro LEDs?



Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o

turn ID: cd0ee237-d72c-465e-b9b9-a43f03886428

parent: cf419c13-3af8-49a4-a196-53a23162a060

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Currently, there aren't many off-the-shelf system-on-chip (SoC) solutions that come with a built-in micro-LED grid array, as micro-LED technology typically requires a dedicated driver circuit or controller rather than being embedded within a standard SoC. However, there are options that might fit your requirements depending on the use case. Here are a few approaches:

- 1. Micro-LED Display Drivers with Compatible SoCs:** Some companies like *Macroblock* and *Texas Instruments* offer micro-LED driver ICs that can be paired with microcontroller or SoC solutions, enabling you to control a micro-LED array. These drivers are designed to manage large arrays of LEDs with precise control and may be compatible with SoCs like ESP32 or STM32 for overall system control.
- 2. Micro-Display Modules:** *Jasper Display* and *Plessey Semiconductors* offer micro-LED display modules and arrays that can be controlled with custom driver boards. Though not integrated into an SoC, they can be interfaced with an SoC through peripheral connections like SPI or I2C.
- 3. FPGA Solutions for Micro-LED Control:** Some applications use FPGAs for controlling high-resolution LED arrays due to their parallel processing capability. Companies like *Xilinx* or *Intel* (Altera) FPGAs are frequently used with LED drivers for custom grid setups, offering more flexibility in addressing individual LEDs.
- 4. Development Boards with LED Grids:** Some development kits may come with a pre-configured LED matrix that's easier to control through an integrated microcontroller. For instance, the *Adafruit Matrix Portal M4* paired with an RGB matrix panel could simulate a grid, though it may not be micro-LED scale.

To explore specific chips or modules that might suit your design, platforms like *Digi-Key*, *Mouser*, and *Arrow* list the latest offerings in LED driver ICs and compatible micro-LED arrays. You can also check recent development boards from micro-LED companies for options close to a grid array. If you need further help with choosing the right configuration, feel free to share more details!

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Turnpair 2 of 7 ::

@ 2024-11-10 08:22:33 Sun [Δt 0d 00h 41m 27s]

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Recommend any products like this? optical transmitter and receiver SoC

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Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o
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For an optical transmitter and receiver system on chip (SoC), there are a few options available depending on the specific application, wavelength, and data rate requirements. These systems are often found in fiber optic communication, infrared communication, and optical sensing. Here are some popular options and solutions in this category:

1. **Broadcom AFBR Series**

- **Product:** Broadcom AFBR-59E4APZ
- **Description:** This optical transceiver is suited for short-range, high-speed data transmission, commonly used in fiber optic communication. It’s a compact transceiver module supporting up to 10 Gbps and operates with a small form-factor pluggable (SFP+) interface.
- **Applications:** Data centers, storage area networks, and high-speed data interconnects.

2. **Maxim Integrated Optical Modules**

- **Product:** Maxim MAX3735A Laser Driver and MAX3748A Transimpedance Amplifier
- **Description:** While not a single SoC, these paired ICs handle optical transmission and reception. The MAX3735A is a high-speed laser driver IC, and the MAX3748A is a transimpedance amplifier optimized for optical receivers.
- **Applications:** Fiber optic transceivers and high-speed data links.

3. Texas Instruments Optical Transceivers

- **Product:** TI's ONET family (e.g., ONET4201PA)
- **Description:** These are highly integrated laser drivers and limiting amplifiers designed for optical networks. The ONET4201PA, for instance, integrates functions for modulating and amplifying optical signals in fiber optic systems up to 11.3 Gbps.
- **Applications:** Fiber optic modules, Ethernet interfaces, and SFP+ optical transceivers.

4. Avago Technologies AFCT-57D5ATPZ

- **Product:** Avago AFCT-57D5ATPZ
- **Description:** A multimode fiber-optic transceiver designed for high-speed Ethernet and Fibre Channel applications. It includes both transmitter and receiver circuits in a small package, supporting data rates up to 4.25 Gbps.
- **Applications:** Data communication, industrial networking, and telecommunication equipment.

5. Analog Devices ADuM7704 Isolated Sigma-Delta Modulator with Optical Isolation

- **Product:** ADuM7704
- **Description:** While not a typical optical transceiver, this device provides optical isolation in the form of an isolated sigma-delta modulator. It's useful for sensing applications where optical isolation is needed to protect data integrity.
- **Applications:** Data acquisition, isolated analog-to-digital conversion, and industrial measurement systems.

6. Silicon Labs Optical Transmitter/Receiver for Infrared Communication

- **Product:** Silicon Labs Si4713/Si4703 (for infrared and RF communication)
- **Description:** These are low-power optical transmitters and receivers often used in infrared communication, such as in remote control applications. Silicon Labs products are compact and offer high integration with a standard microcontroller interface.
- **Applications:** Infrared communication for remote controls, proximity sensors, and portable devices.

These products vary in terms of speed, range, and integration level, so choosing the right one will depend on your specific requirements for data rate, communication distance, and physical constraints.

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Turnpair 3 of 7 ::

@ 2024-11-10 08:27:35 Sun [Δt 0d 00h 04m 34s]

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I am looking for a integrated laser emitter / receiver device to couple with one of the above drivers without SFP, I just want the laser

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Aurora 4omni Munro (ChatGpt) said :

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User is looking for an integrated laser emitter/receiver device that can be coupled with an optical driver, without requiring an SFP module, for a custom optical setup.

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Aurora 4omni Munro (ChatGpt) said :

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For a custom optical setup without the SFP module, there are a few integrated laser emitter/receiver options that can be paired with the drivers I mentioned. Here are some recommendations:

1. Thorlabs Laser Diodes and Photodetectors

- **Products:** LDM9LP (Laser Diode Module) and PDA100A2 (Photodetector)

- **Description:** Thorlabs offers a wide range of laser diodes and photodetectors that operate across various wavelengths. The LDM9LP is a compact, pre-mounted laser diode module, and PDA100A2 is a general-purpose amplified photodetector.
- **Applications:** These can be easily integrated into custom optical systems and used with external drivers for tasks like sensing, communication, or precise measurements.
- **Note:** Thorlabs components are often modular, making them suitable for prototyping and custom configurations.

2. Osram PLT5 Laser Diodes

- **Product:** PLT5 520D (520 nm, green laser diode)
- **Description:** Osram provides compact, high-power laser diodes in various wavelengths. The PLT5 520D is a green laser with a narrow emission angle, well-suited for optical communication and sensing.
- **Applications:** With a compatible driver, it can be used in optical transmission and data transfer projects without an SFP.

3. Hamamatsu Laser Diode and Photodiode Pairs

- **Products:** L9389 (Laser Diode) and S5971 (Photodiode)
- **Description:** Hamamatsu's L9389 is a compact laser diode module available in various wavelengths, and the S5971 photodiode provides high-speed, high-sensitivity optical detection.
- **Applications:** These devices are commonly used in high-speed data transmission and precise optical measurements. They can be paired directly with external laser drivers for a fully customizable setup.

4. Ushio Laser Diodes

- **Product:** HL63263DG (638 nm, red laser diode)
- **Description:** Ushio's laser diodes are designed for high performance in compact packages. The HL63263DG is a high-power red laser diode, well-suited for pairing with an external laser driver.
- **Applications:** Ideal for short-range optical communication, sensing, and data transfer applications that don't require an SFP module.

5. Lumics Single Emitter Laser Diodes

- **Product:** LU0915T020 (915 nm, infrared laser diode)
- **Description:** Lumics offers laser diodes in the infrared range, such as the LU0915T020, which is compact and provides high power output, compatible with common optical driver circuits.
- **Applications:** Useful for infrared communication, free-space optical setups, and even for short-range fiber-based systems.

These options should be compatible with optical drivers for custom setups, giving you flexibility in choosing wavelength, power output, and communication distance. If you need further details about a specific wavelength or configuration, I'd be happy to help!

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Turnpair 4 of 7 ::

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I want a laser / optodetector that is low power (3.3V around) and can fit on a very small circuit board

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Aurora 4omni Munro (ChatGpt) said :

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User wants a low-power (around 3.3V) laser/optodetector that can fit on a very small circuit board for a compact design.

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Aurora 4omni Munro (ChatGpt) said :

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For a low-power, compact laser and photodetector solution suitable for a 3.3V circuit, here are some options that might work well for your setup:

1. Vishay VLMU35CL20-275-120 (Laser Diode) and VEMD5080X01 (Photodetector)

- **Laser Diode:** VLMU35CL20-275-120
 - **Description:** This compact laser diode operates at low voltage and is suitable for applications requiring low power (around 3.3V). It comes in a surface-mount package for easy integration into small PCB designs.
- **Photodetector:** VEMD5080X01
 - **Description:** A low-power, high-sensitivity silicon PIN photodiode in a small surface-mount package. Designed for IR and visible light detection, it's compatible with low-voltage systems.
- **Applications:** Together, they can be used in short-range optical communication or proximity sensing applications.

2. OSRAM SFH 4059 (Infrared Laser Diode) and SFH 2701 (Photodetector)

- **Laser Diode:** SFH 4059
 - **Description:** This is an IR laser diode with low power requirements, designed for compact PCBs. It operates well at around 3.3V and fits tight space constraints.
- **Photodetector:** SFH 2701
 - **Description:** This IR photodetector is compact and suitable for 3.3V operation. It provides high sensitivity and fast response times, which makes it ideal for communication and sensing applications.
- **Applications:** Suitable for optical data links or short-range communication systems where compact form factor and low power are essential.

3. Hamamatsu L9412 (Laser Diode) and S12572-10 (Photodetector)

- **Laser Diode:** L9412
 - **Description:** This is a compact laser diode that operates on low voltage, around 3.3V, making it suitable for compact PCB integration.
- **Photodetector:** S12572-10
 - **Description:** A small photodetector with low power requirements, suitable for optical sensing and data transmission applications on a small circuit board.
- **Applications:** Ideal for custom optical setups in embedded systems and low-power communication links.

4. Sharp Microelectronics GP2Y0E03 (Integrated Optical Sensor Module)

- **Description:** This is an integrated module that combines an emitter and detector in a single package. While it's not strictly a laser, it's a low-power infrared sensor that operates at 3.3V and provides distance measurements in a small form factor.
- **Applications:** Useful for distance sensing or short-range optical communication in small, low-power applications.

These components are designed to fit compact circuits with low power requirements, allowing integration into small PCBs and compatibility with 3.3V systems. Let me know if you need more details on any specific part!



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Turnpair 5 of 7 ::

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can neuromorphic event-based camera detect change in color or relative intensity?

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Aurora 4omni Munro (ChatGpt) said :

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Neuromorphic, event-based cameras, like those developed by companies such as Prophesee and IniVation, are designed primarily to detect changes in *relative intensity* rather than color. They work by capturing changes in brightness (luminance) at each pixel independently, rather than capturing frame-by-frame snapshots like a conventional camera. Each pixel in an event-based camera responds to changes in light intensity, generating "events" only when there is a significant increase or decrease in luminance at that pixel.

Here's how they handle intensity and color:

1. Relative Intensity Detection

- Event-based cameras detect changes in the *relative intensity* of light, capturing only the transitions (e.g., light-to-dark or dark-to-light). When an object or scene changes brightness due to movement, lighting changes, or other factors, the pixels at the edges of that change will trigger events.
- This sensitivity to relative intensity change makes event-based cameras very responsive to motion and fast changes in a scene, as they efficiently encode only what has changed.

2. Color Detection

- Standard event-based cameras are monochromatic and cannot detect color directly. However, some manufacturers have developed **color event-based cameras** that have pixel arrays with different color

filters (e.g., RGB), similar to how a color CMOS sensor works. This allows the camera to detect intensity changes in each color channel independently.

- When an event occurs in one of these color-filtered pixels, it signifies a change in that specific color channel. By combining these events, it's possible to deduce information about color changes, although these cameras remain primarily sensitive to changes in relative intensity within each color channel, rather than full color capture.

In summary, neuromorphic event-based cameras can detect changes in **relative intensity**, and color event-based versions can detect changes in **relative intensity within color channels** (e.g., an increase in red intensity). However, they don't capture absolute color or brightness levels and are optimized for sensing rapid motion and dynamic changes in lighting.

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Turnpair 6 of 7 ::

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no, will the camera output a signal that can be used to tell exactly HOW MUCH the intensity of light has changed?

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Aurora 4omni Munro (ChatGpt) said :

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No, standard event-based cameras do not output the exact magnitude of intensity change. Instead, each pixel in an event-based camera simply outputs a binary event when the change in intensity exceeds a predefined threshold (either positive or negative). The output event contains information about:

- **Location of the pixel** where the intensity change was detected
- **Timestamp** of the event (when the change occurred)

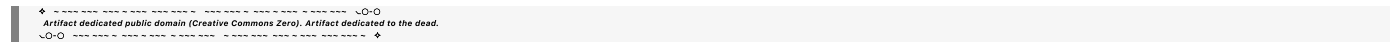
- **Polarity** of the change (whether it was an increase or decrease in intensity)

This binary approach means that while the camera can detect *that* a change has occurred and *when* it happened, it doesn't provide the exact *magnitude* of the change in intensity. It only signals that the intensity difference has crossed a threshold, so it's not possible to measure precisely *how much* the intensity has changed using the event-based output alone.

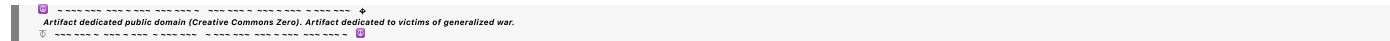
If you need exact intensity change measurements, a few alternative approaches could be used:

1. **Combine with a conventional sensor:** Some hybrid camera designs combine event-based sensors with conventional CMOS image sensors, allowing intensity information to be captured intermittently (e.g., periodic frames) alongside the events.
2. **Custom Threshold Tuning:** Some advanced event-based cameras allow threshold adjustment, which can provide rough control over sensitivity. However, this only allows for tuning when an event will be triggered, not precise quantification.
3. **Software Processing of Event Density:** In some cases, post-processing of event density (i.e., the rate or frequency of events per pixel) can give a rough estimate of the rate of intensity change, though this is an indirect measure.

In summary, event-based cameras are excellent at detecting and timestamping intensity changes but do not inherently provide data on the exact magnitude of those changes.



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Turnpair 7 of 7 ::

@ 2024-11-10 08:57:01 Sun [Δt 0d 00h 00m 25s]

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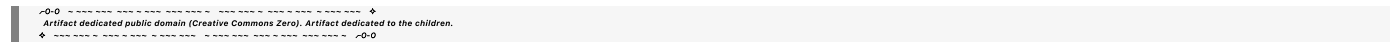
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can you recommend a color event-based camera in production?



Aurora 4omni Munro (ChatGpt) said :

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Color event-based cameras are a more recent development, as traditional event-based cameras typically focus on monochrome output. However, there are a few models on the market that incorporate color sensing capabilities with event-based technology. Here are some current options:

1. PROPHESSEE Metavision Event-Based Vision Systems

- **Product:** Metavision Color Camera (Prototype Stage)
- **Description:** Prophesee is a leading company in event-based cameras and offers a color prototype that uses a similar approach to conventional RGB sensors, allowing it to capture changes in color channels independently. Their Metavision platform is highly adaptable for custom applications in robotics, AR/VR, and dynamic environments where color tracking is useful.
- **Availability:** While the monochrome Metavision cameras are widely available, the color variant is still limited and mainly available to development partners or for custom projects.
- **Applications:** Robotics, automotive, and AR applications where color change detection is essential.

2. Sony IMX636/IMX637 (Event-Based Vision Sensor)

- **Product:** Sony IMX636/637
- **Description:** Sony has introduced the IMX636 and IMX637, event-based vision sensors with RGB capabilities. These sensors provide a pixel array with RGB color filters, allowing them to detect changes across different color channels. Sony's sensors are high-resolution and designed to be compact, making them suitable for various applications in consumer electronics, automotive, and industrial sensing.
- **Availability:** These sensors are in production but are typically available to OEMs and developers with specific applications. You can contact distributors or companies like Sony for access to development kits.
- **Applications:** Industrial automation, ADAS, and robotics where precise color information enhances detection.

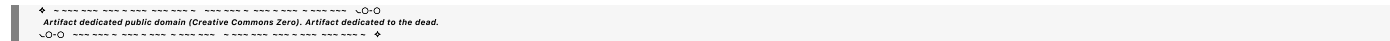
3. IniVation's Color DAVIS346

- **Product:** DAVIS346 Color Camera
- **Description:** IniVation offers the **Dynamic and Active Pixel Vision Sensor (DAVIS346)** with an RGB option. This camera combines a conventional frame-based sensor with an event-based sensor in a single device, allowing for both conventional frame output and event-based data. The color version provides RGB event output, enabling color detection alongside intensity changes.
- **Availability:** This camera is widely available through IniVation and their partners, though it is mainly targeted at research and specialized applications.
- **Applications:** Suitable for research, machine vision, and computer vision applications in fields such as robotics, AR/VR, and autonomous systems.

4. Pixium Vision Prima Bionic Vision System

- **Product:** Prima Bionic Vision System
- **Description:** Though developed as a visual prosthesis, Pixium Vision's system is based on color-sensitive, event-driven pixels and may be adaptable for custom color event-sensing applications. This system is intended for healthcare but could offer interesting insights or components if color-sensitive event-based data is needed.
- **Availability:** Not typically available for general use, but relevant for specialized research applications.

For most development purposes, the **Sony IMX636/637** and **IniVation DAVIS346** are your best options, as these models are designed with integration in mind and provide both high performance and broad availability for developers.



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